```sql
    @Nickname NVARCHAR(50),
    @RealName NVARCHAR(100),
    @Country NVARCHAR(50),
    @BirthDate DATE,
    @TeamId INT = NULL,
    @NewPlayerId INT OUTPUT
AS
BEGIN
    SET NOCOUNT ON;

    IF DATEADD(YEAR, 16, @BirthDate) > CAST(GETDATE() AS DATE)
        THROW 50001, N'Игроку меньше 16 лет, регистрация запрещена', 1;

    IF @TeamId IS NOT NULL
    BEGIN
        IF NOT EXISTS (SELECT 1 FROM dbo.Teams WHERE TeamId = @TeamId)
            THROW 50002, N'Команда с таким id не существует', 1;
        IF NOT EXISTS (SELECT 1 FROM dbo.Teams WHERE TeamId = @TeamId AND IsActive = 1)
            THROW 50003, N'Команда неактивна, принимать игроков нельзя', 1;
    END

    INSERT INTO dbo.Players (Nickname, RealName, Country, BirthDate, TeamId)
    VALUES (@Nickname, @RealName, @Country, @BirthDate, @TeamId);

    SET @NewPlayerId = SCOPE_IDENTITY();
END;
GO
```

68 % Проблемы не найдены.

Сообщения

Выполнение команд успешно завершено.

Время выполнения: 2026-09-23T10:13:37.9486238+02:00

```sql
CREATE OR ALTER PROCEDURE dbo.CompleteMatch
    @MatchId INT,
    @Team1Score INT,
    @Team2Score INT
AS
BEGIN
    SET NOCOUNT ON;
    SET XACT_ABORT ON;

    BEGIN TRY
        BEGIN TRAN;

        IF NOT EXISTS (SELECT 1 FROM dbo.Matches WHERE MatchId = @MatchId)
            THROW 50010, N'Матч не найден', 1;

        IF EXISTS (SELECT 1 FROM dbo.Matches WHERE MatchId = @MatchId AND IsCompleted = 1)
            THROW 50011, N'Матч уже завершён', 1;

        IF @Team1Score = @Team2Score
            THROW 50012, N'Ничья недопустима', 1;

        DECLARE @WinnerId INT;
        SELECT @WinnerId = CASE WHEN @Team1Score > @Team2Score THEN Team1Id ELSE Team2Id END
        FROM dbo.Matches
        WHERE MatchId = @MatchId;

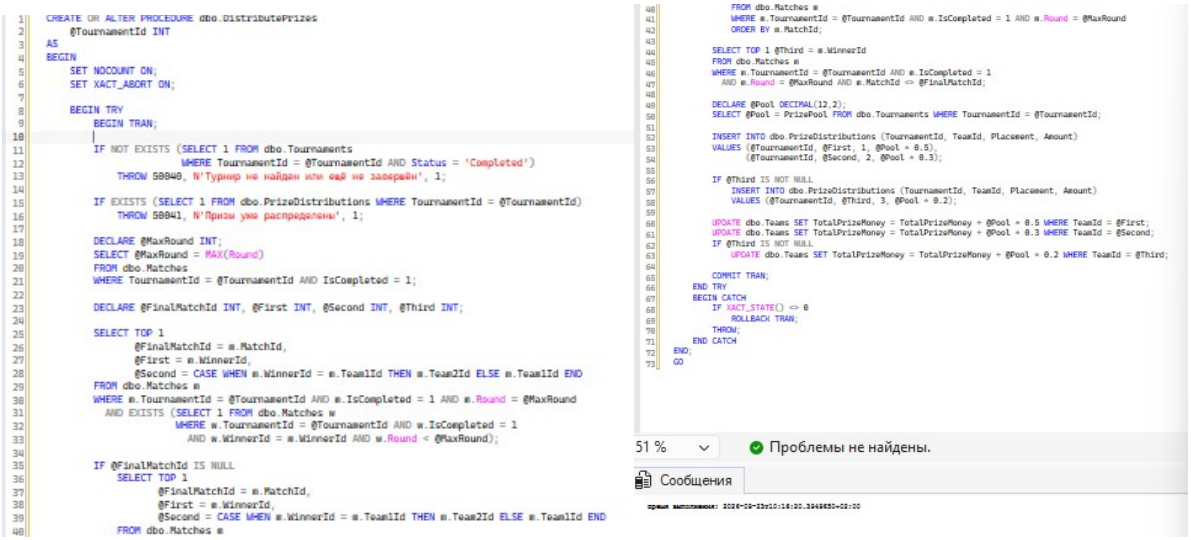
        UPDATE dbo.Matches
        SET Team1Score = @Team1Score,
            Team2Score = @Team2Score,
            WinnerId = @WinnerId,
            IsCompleted = 1
        WHERE MatchId = @MatchId;

        COMMIT TRAN;
    END TRY
    BEGIN CATCH
        IF XACT_STATE() <> 0
            ROLLBACK TRAN;
        THROW;
    END CATCH
END;
GO
```

47 % Проблемы не найдены.

Сообщения

Время выполнения: 2026-09-23T10:13:38.2233683+02:00



```sql
CREATE OR ALTER PROCEDURE dbo.DistributePrizes
    @TournamentId INT
AS
BEGIN
    SET NOCOUNT ON;
    SET XACT_ABORT ON;

    BEGIN TRY
        BEGIN TRAN;

        IF NOT EXISTS (SELECT 1 FROM dbo.Tournaments
                       WHERE TournamentId = @TournamentId AND Status = 'Completed')
            THROW 50040, N'Турнир не найден или ещё не завершён', 1;

        IF EXISTS (SELECT 1 FROM dbo.PrizeDistributions WHERE TournamentId = @TournamentId)
            THROW 50041, N'Призы уже распределены', 1;

        DECLARE @MaxRound INT;
        SELECT @MaxRound = MAX(Round)
        FROM dbo.Matches
        WHERE TournamentId = @TournamentId AND IsCompleted = 1;

        DECLARE @FinalMatchId INT, @First INT, @Second INT, @Third INT;

        SELECT TOP 1
               @FinalMatchId = m.MatchId,
               @First = m.WinnerId,
               @Second = CASE WHEN m.WinnerId = m.Team1Id THEN m.Team2Id ELSE m.Team1Id END
        FROM dbo.Matches m
        WHERE m.TournamentId = @TournamentId AND m.IsCompleted = 1 AND m.Round = @MaxRound
          AND EXISTS (SELECT 1 FROM dbo.Matches w
                      WHERE w.TournamentId = @TournamentId AND w.IsCompleted = 1
                        AND w.WinnerId = m.WinnerId AND w.Round < @MaxRound);

        IF @FinalMatchId IS NULL
            SELECT TOP 1
                   @FinalMatchId = m.MatchId,
                   @First = m.WinnerId,
                   @Second = CASE WHEN m.WinnerId = m.Team1Id THEN m.Team2Id ELSE m.Team1Id END
              FROM dbo.Matches m
              WHERE m.TournamentId = @TournamentId AND m.IsCompleted = 1 AND m.Round = @MaxRound
              ORDER BY m.MatchId;

        SELECT TOP 1 @Third = m.WinnerId
        FROM dbo.Matches m
        WHERE m.TournamentId = @TournamentId AND m.IsCompleted = 1
          AND m.Round = @MaxRound AND m.MatchId <> @FinalMatchId;

        DECLARE @Pool DECIMAL(12,2);
        SELECT @Pool = PrizePool FROM dbo.Tournaments WHERE TournamentId = @TournamentId;

        INSERT INTO dbo.PrizeDistributions (TournamentId, TeamId, Placement, Amount)
        VALUES (@TournamentId, @First, 1, @Pool * 0.5),
               (@TournamentId, @Second, 2, @Pool * 0.3);

        IF @Third IS NOT NULL
            INSERT INTO dbo.PrizeDistributions (TournamentId, TeamId, Placement, Amount)
            VALUES (@TournamentId, @Third, 3, @Pool * 0.2);

        UPDATE dbo.Teams SET TotalPrizeMoney = TotalPrizeMoney + @Pool * 0.5 WHERE TeamId = @First;
        UPDATE dbo.Teams SET TotalPrizeMoney = TotalPrizeMoney + @Pool * 0.3 WHERE TeamId = @Second;
        IF @Third IS NOT NULL
            UPDATE dbo.Teams SET TotalPrizeMoney = TotalPrizeMoney + @Pool * 0.2 WHERE TeamId = @Third;

        COMMIT TRAN;
    END TRY
    BEGIN CATCH
        IF XACT_STATE() <> 0
            ROLLBACK TRAN;
        THROW;
    END CATCH
END;
GO
```

51 % Проблемы не найдены.

Сообщения

время выполнения: 2026-09-23T10:16:20.3949650+03:00

```sql
CREATE OR ALTER FUNCTION dbo.GetPlayerKDA (@PlayerId INT)
RETURNS DECIMAL(10,2)
AS
BEGIN
    DECLARE @KDA DECIMAL(10,2);

    SELECT @KDA = AVG(CAST(Kills + Assists AS DECIMAL(10,2)) / CASE WHEN Deaths = 0 THEN 1 ELSE Deaths END)
    FROM dbo.PlayerMatchStats
    WHERE PlayerId = @PlayerId;

    RETURN @KDA;
END;
GO
```

51 % Проблемы не найдены.

Сообщения

выполнение команд успешно завершено.

время выполнения: 2026-09-23T10:16:45.9432770+03:00

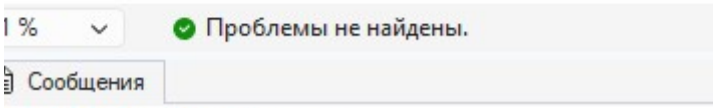
```sql
CREATE OR ALTER FUNCTION dbo.GetTeamMatchHistory (@TeamId INT, @GamesBack INT = NULL)
RETURNS TABLE
AS
RETURN
(
    SELECT TOP (ISNULL(@GamesBack, 2147483647))
        m.MatchId,
        m.Round,
        CASE WHEN m.Team1Id = @TeamId THEN t2.Name ELSE t1.Name END AS OpponentName,
        CASE WHEN m.Team1Id = @TeamId THEN m.Team1Score ELSE m.Team2Score END AS TeamScore,
        CASE WHEN m.Team1Id = @TeamId THEN m.Team2Score ELSE m.Team1Score END AS OpponentScore,
        CASE
            WHEN m.IsCompleted = 0 THEN N'Pending'
            WHEN m.WinnerId = @TeamId THEN N'Win'
            ELSE N'Loss'
        END AS Result,
        m.MatchDate
    FROM dbo.Matches m
    JOIN dbo.Teams t1 ON t1.TeamId = m.Team1Id
    JOIN dbo.Teams t2 ON t2.TeamId = m.Team2Id
    WHERE m.Team1Id = @TeamId OR m.Team2Id = @TeamId
    ORDER BY m.MatchDate DESC, m.MatchId DESC
);
```

1 % Проблемы не найдены.

Сообщения

время выполнения: 2026-09-23T10:58:48.4508708+03:00

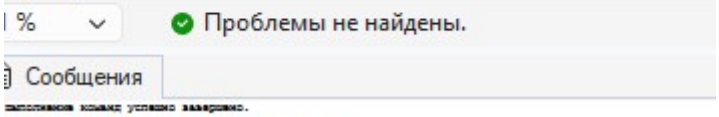
```sql
CREATE OR ALTER FUNCTION dbo.GetTeamPowerRating (@TeamId INT)
RETURNS DECIMAL(10,2)
WITH SCHEMABINDING
AS
BEGIN
    DECLARE @Completed INT, @Wins INT, @WinRate DECIMAL(5,2);
    DECLARE @Prize DECIMAL(12,2), @Sponsor DECIMAL(12,2), @Rating DECIMAL(10,2);
    SELECT @Completed = COUNT(*),
           @Wins = SUM(CASE WHEN m.WinnerId = @TeamId THEN 1 ELSE 0 END)
    FROM dbo.Matches AS m
    WHERE m.IsCompleted = 1 AND (m.Team1Id = @TeamId OR m.Team2Id = @TeamId);
    SET @WinRate = ISNULL(CAST(@Wins AS DECIMAL(5,2)) / NULLIF(@Completed, 0), 0);
    SELECT @Prize = TotalPrizeMoney FROM dbo.Teams WHERE TeamId = @TeamId;
    SELECT @Sponsor = ISNULL(SUM(ContributionAmount), 0)
    FROM dbo.TeamSponsors WHERE TeamId = @TeamId;
    SET @Rating = @WinRate * 100 + (@Prize + @Sponsor) / 10000.0;
    RETURN @Rating;
END;
GO
```




```
Выполнение команд успешно завершено.

Время выполнения: 2026-05-23T12:00:04.8206355+03:00
```

```sql
CREATE OR ALTER TRIGGER dbo.trg_Teams_PreventDeleteActive
ON dbo.Teams
AFTER DELETE
AS
BEGIN
    SET NOCOUNT ON;
    IF EXISTS (SELECT 1 FROM deleted WHERE IsActive = 1)
    BEGIN
        ROLLBACK TRANSACTION;
        THROW 50020, N'Нельзя удалить активную команду. Сначала деактивируйте её!', 1;
    END
END;
```



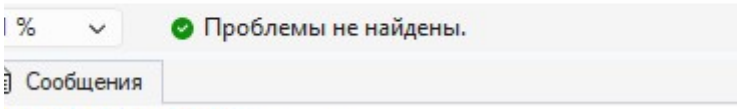

```
Выполнение команд успешно завершено.

Время выполнения: 2026-05-23T12:00:26.2100287+03:00
```

```sql
CREATE OR ALTER TRIGGER dbo.trg_Matches_CreateStatsStubs
ON dbo.Matches
AFTER UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT UPDATE(IsCompleted)
        RETURN;

    INSERT INTO dbo.PlayerMatchStats (PlayerId, MatchId, Kills, Deaths, Assists, IsMVP)
    SELECT p.PlayerId, i.MatchId, 0, 0, 0, 0
    FROM inserted i
    JOIN deleted d ON d.MatchId = i.MatchId
    JOIN dbo.Players p ON p.TeamId IN (i.Team1Id, i.Team2Id)
    WHERE d.IsCompleted = 0 AND i.IsCompleted = 1
      AND NOT EXISTS (SELECT 1 FROM dbo.PlayerMatchStats s
                      WHERE s.PlayerId = p.PlayerId AND s.MatchId = i.MatchId);
END;
GO
```




```
Выполнение команд успешно завершено.

Время выполнения: 2026-05-23T11:11:28.6069628+03:00
```

```sql
CREATE OR ALTER VIEW dbo.vw_PlayerTransfers
AS
SELECT p.PlayerId, p.Nickname, p.TeamId, t.Name AS TeamName
FROM dbo.Players p
LEFT JOIN dbo.Teams t ON t.TeamId = p.TeamId;
GO

CREATE OR ALTER TRIGGER dbo.trg_vwPlayerTransfers_Transfer
ON dbo.vw_PlayerTransfers
INSTEAD OF UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT UPDATE(TeamId)
        RETURN;

    DECLARE @PlayerId INT, @OldTeamId INT, @NewTeamId INT;
    SELECT @PlayerId = i.PlayerId, @OldTeamId = d.TeamId, @NewTeamId = i.TeamId
    FROM inserted i
    JOIN deleted d ON d.PlayerId = i.PlayerId;

    IF NOT EXISTS (SELECT 1 FROM dbo.Teams WHERE TeamId = @NewTeamId AND IsActive = 1)
    BEGIN
        ROLLBACK TRANSACTION;
        THROW 50031, N'Новая команда не существует или неактивна', 1;
    END

    IF (SELECT COUNT(*) FROM dbo.Players WHERE TeamId = @NewTeamId) >= 5
    BEGIN
        ROLLBACK TRANSACTION;
        THROW 50032, N'Состав заполнен', 1;
    END

    IF @NewTeamId = @OldTeamId
    BEGIN
        ROLLBACK TRANSACTION;
        THROW 50030, N'Игрок уже состоит в этой команде', 1;
    END

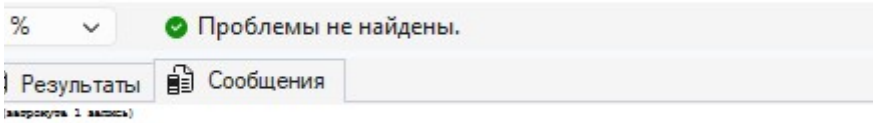
    UPDATE dbo.Players SET TeamId = @NewTeamId WHERE PlayerId = @PlayerId;
END;
GO
```

```sql
--регистрация в команду
DECLARE @NewId INT;
EXEC dbo.RegisterPlayer @Nickname = N'NewStar', @RealName = N'Пётр Новиков',
     @Country = N'Россия', @BirthDate = '2005-03-10', @TeamId = 2, @NewPlayerId = @NewId OUTPUT;
SELECT @NewId AS NewPlayerId;

--регистрация без команды
DECLARE @FreeId INT;
EXEC dbo.RegisterPlayer @Nickname = N'FreeAgent', @RealName = N'Ли Сону',
     @Country = N'Южная Корея', @BirthDate = '2004-01-11', @TeamId = NULL, @NewPlayerId = @FreeId OUTPUT;
SELECT @FreeId AS FreeAgentId;

--игроку 11 лет
DECLARE @KidId INT;
EXEC dbo.RegisterPlayer @Nickname = N'Kid', @RealName = N'Мальчик Мальчиков',
     @Country = N'Россия', @BirthDate = '2015-01-01', @TeamId = NULL, @NewPlayerId = @KidId OUTPUT;

--команда 5 неактивна
DECLARE @BadId INT;
EXEC dbo.RegisterPlayer @Nickname = N'Unlucky', @RealName = N'Тест Тестов',
     @Country = N'США', @BirthDate = '2000-01-01', @TeamId = 5, @NewPlayerId = @BadId OUTPUT;
GO
```



(затронута 1 запись)

(затронута 1 запись)
сообщение: 50001, уровень: 16, состояние: 1, процедура: dbo.registerplayer, строка: 13 [строка начала пакета: 0]
игроку меньше 16 лет, регистрация запрещена

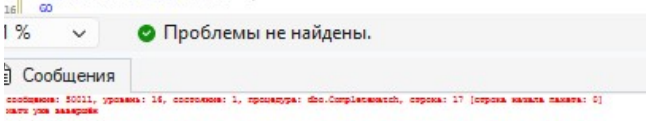
время выполнения: 2026-05-23T11:17:13.9997634+03:00

```sql
--завершение матча 3, счёт 2:1 в пользу команды 1
EXEC dbo.CompleteMatch @MatchId = 3, @Team1Score = 2, @Team2Score = 1;
SELECT MatchId, Team1Score, Team2Score, WinnerId, IsCompleted
FROM dbo.Matches WHERE MatchId = 3;

--начнём на матча 4
EXEC dbo.CompleteMatch @MatchId = 4, @Team1Score = 1, @Team2Score = 1;

--повторное завершения матча 3
EXEC dbo.CompleteMatch @MatchId = 3, @Team1Score = 2, @Team2Score = 0;

--завершение матча 4, счёт 0:2 в пользу команды 2
EXEC dbo.CompleteMatch @MatchId = 4, @Team1Score = 0, @Team2Score = 2;
SELECT MatchId, Team1Score, Team2Score, WinnerId, IsCompleted
FROM dbo.Matches WHERE MatchId = 4;
GO
```



сообщение: 50111, уровень: 16, состояние: 1, процедура: dbo.CompleteMatch, строка: 17 [строка начала пакета: 0]
матч уже завершён

время выполнения: 2026-05-23T11:27:31.9404093+03:00

```sql
--KDA одного игрока
SELECT dbo.GetPlayerKDA(1) AS KDA_Fade;

--KDA всех игроков команды 1
SELECT p.Nickname, dbo.GetPlayerKDA(p.PlayerId) AS KDA
FROM dbo.Players p
WHERE p.TeamId = 1;

--вся история матчей команды 4
SELECT * FROM dbo.GetTeamMatchHistory(4, NULL);

--только последний матч команды 4
SELECT * FROM dbo.GetTeamMatchHistory(4, 1);

--рейтинг всех активных команд по убыванию
SELECT t.Name, dbo.GetTeamPowerRating(t.TeamId) AS PowerRating
FROM dbo.Teams t
WHERE t.IsActive = 1
ORDER BY PowerRating DESC;
GO
```

	KDA_Fade
1	3.67

	Nickname	KDA
1	Fade	3.67
2	ShadowBlade	2.00
3	Lumin	4.75
4	Kraken	2.13
5	Hexen	2.83

	MatchId	Round	OpponentName	TeamScore	OpponentScore	Result	MatchDate
1	3	2	Nexus Gaming	1	2	Loss	2025-12-05
2	2	1	Phoenix Reign	2	1	Win	2025-12-02

	MatchId	Round	OpponentName	TeamScore	OpponentScore	Result	MatchDate
1	3	2	Nexus Gaming	1	2	Loss	2025-12-05

	Name	PowerRating
1	Nexus Gaming	123.00
2	Silent Storm	114.00
3	Phoenix Reign	40.50
4	Void Hunters	11.50

```sql
--турнир завершён, можно раздавать призы
UPDATE dbo.Tournaments SET Status = 'Completed' WHERE TournamentId = 1;
GO
EXEC dbo.DistributePrizes @TournamentId = 1;
GO
--кому сколько досталось
SELECT pd.Placement, t.Name, pd.Amount
FROM dbo.PrizeDistributions pd
JOIN dbo.Teams t ON t.TeamId = pd.TeamId
WHERE pd.TournamentId = 1
ORDER BY pd.Placement;

--как выросли суммарные призовые команд
SELECT TeamId, Name, TotalPrizeMoney FROM dbo.Teams ORDER BY TeamId;

--ошибка повторной раздачи
EXEC dbo.DistributePrizes @TournamentId = 1;
GO
```

1 % Проблемы не найдены.

Результаты Сообщения

```
(затронута 1 запись)

(затронуто записей: 2)

(затронуто записей: 5)
сообщение: 50041, уровень: 16, состояние: 1, процедура: dbo.distributeprizes, строка: 16 [строка начала пакета: 5]
призы уже распределены

время выполнения: 2026-03-23T11:33:10.0864583+03:00
```

```sql
--заглушки создались сами
SELECT MatchId, PlayerId, Kills, Deaths, Assists, IsMVP
FROM dbo.PlayerMatchStats
WHERE MatchId IN (3, 4)
ORDER BY MatchId, PlayerId;    --20 нулевых строк, по 10 на матч

--счёт уже завершённого матча, триггер должен промолчать
UPDATE dbo.Matches SET Team1Score = 3 WHERE MatchId = 3;
SELECT COUNT(*) AS StubCountMatch3 FROM dbo.PlayerMatchStats WHERE MatchId = 3;    -- всё ещё 10
GO
```

51 % Проблемы не найдены.

Результаты Сообщения

	MatchId	PlayerId	Kills	Deaths	Assists	IsMVP
1	3	1	0	0	0	0
2	3	2	0	0	0	0
3	3	3	0	0	0	0
4	3	4	0	0	0	0
5	3	5	0	0	0	0
6	3	16	0	0	0	0
7	3	17	0	0	0	0
8	3	18	0	0	0	0

	StubCountMatch3
1	10

```sql
--UPDATE через представление
--место в Void Hunters, чтобы успешный трансфер был возможен
UPDATE dbo.Players SET TeamId = NULL WHERE PlayerId = 15;

--Fade из Nexus Gaming в Void Hunters
UPDATE dbo.vw_PlayerTransfers SET TeamId = 3 WHERE PlayerId = 1;
SELECT PlayerId, Nickname, TeamId, TeamName FROM dbo.vw_PlayerTransfers WHERE PlayerId = 1;

--в Silent Storm уже пять игроков
UPDATE dbo.vw_PlayerTransfers SET TeamId = 4 WHERE PlayerId = 1;

--Echo Squad выключена
UPDATE dbo.vw_PlayerTransfers SET TeamId = 5 WHERE PlayerId = 2;

--переход в свою же команду
UPDATE dbo.vw_PlayerTransfers SET TeamId = 3 WHERE PlayerId = 1;
GO
```

51 % Проблемы не найдены.

Результаты Сообщения

(затронута 1 запись)
сообщение: 50232, уровень: 16, состояние: 1, процедура: trg_vwPlayerTransfers_InsteadOf, строка: 30 [строка начала пакета: 0]
Состав заполнен

время выполнения: 2026-09-23T11:37:32.7066070+03:00

```sql
--DELETE
--команда для эксперимента
INSERT INTO dbo.Teams (Name, Country, FoundedDate, IsActive)
VALUES (N'Demo Team', N'Китай', '2024-01-01', 1);

--команда активна, удаление запрещено
DELETE FROM dbo.Teams WHERE Name = N'Demo Team';

UPDATE dbo.Teams SET IsActive = 0 WHERE Name = N'Demo Team';
DELETE FROM dbo.Teams WHERE Name = N'Demo Team';
SELECT COUNT(*) AS DemoTeamLeft FROM dbo.Teams WHERE Name = N'Demo Team';   -- ждём 0
GO
```

51 % Проблемы не найдены.

Результаты Сообщения

	DemoTeamLeft
1	0